ERC-4626 Yield Vault with MCDM Scoring

Sergei Solovev
sesesolovev@edu.hse.ru
HSE University, April 2026

Project 1: Custom DeFi Protocol

Why This Project?

The problem with DeFi yields

Lending protocols like Aave and Compound offer variable interest rates that change every block. Users who want the best yield face three challenges:

- **Monitoring** -- rates shift every ~12 seconds, impossible to track manually
- **Gas costs** -- each rebalance costs $2-50, eating into profits
- **Risk** -- chasing the highest APY ignores utilization risk and rate stability

The idea: an autonomous agent

What if a **software agent** could watch the rates, evaluate multiple risk factors, and move funds automatically -- while proving every decision cryptographically on-chain?

This is **agentic DeFi**: an off-chain agent that thinks, and an on-chain vault that verifies and executes.

How It Works

User deposits USDC into the vault and receives `aiUSDC` shares.
From that point, the agent manages everything automatically.

```
+----------------------------------------------------------+
|  Off-chain: Python Agent (runs every hour)               |
|  1. Read APY, utilization, TVL, gas price                |
|  2. Smooth rates with EMA (anti-manipulation)            |
|  3. Score protocols with MCDM (4 factors)                |
|  4. Sign decision with EIP-712 typed data                |
+-----------------------------+----------------------------+
                              | signed tx
+-----------------------------+----------------------------+
|  On-chain: AIVault.sol (ERC-4626 + UUPS proxy)          |
|  - Verify keeper signature (ECDSA)                       |
|  - Check nonce, timestamp, cooldown                      |
|  - Execute rebalance via adapter                         |
|  +-- AaveV3Adapter    +-- CompoundV3Adapter              |
+----------------------------------------------------------+
  Fallback: Chainlink Automation (if agent offline > 6h)
```

Key insight: The agent can be complex (multi-factor analysis), but the vault only trusts cryptographic proofs — not the agent itself.

Key Formulas

ERC-4626 share price (with inflation attack protection):

$$s = \left\lfloor \frac{a \cdot (S + 10^6)}{A + 1} \right\rfloor$$

APY normalization (cross-protocol, to annual 1e18 scale):

- Aave V3: $\text{APY} = \text{liquidityRate}_{RAY} / 10^9$
- Compound V3: $\text{APY} = r_{sec} \times 31{,}557{,}600$

EMA smoothing (dampens noise): $S_t = 0.3 \cdot R_t + 0.7 \cdot S_{t-1}$

MCDM scoring model:

$$\text{Score}_i = 0.40 \cdot f_{APY} + 0.25 \cdot f_{Risk} + 0.20 \cdot f_{Cost} + 0.15 \cdot f_{Stability}$$

Rebalance if: $\text{Score}_{best} - \text{Score}_{current} \geq 0.05$

Example: Aave offers 6% APY but has 95% utilization (risky). Compound offers 5% but with 30% utilization (safe). Simple APY comparison picks Aave. Our MCDM model picks Compound -- the safer choice.

Security and Testing

7 threat mitigations:

Threat	Protection
Inflation attack	Virtual shares (10^6 offset)
Reentrancy	ReentrancyGuard on all external functions
Rate manipulation	EMA + 5% jump guard
Signature forgery	EIP-712 domain + ECDSA verification
Replay attack	Sequential nonce + 5-min timestamp TTL
Agent downtime	Chainlink Automation fallback (6h)
Rapid exploitation	1-hour cooldown between rebalances

67 tests, 0 failures:

Category	Tests	Method
Unit (Solidity)	37	Concrete + fuzz (1000 runs each)
Integration	4	Full lifecycle (deposit -> rebalance -> yield -> withdraw)

Deployment

Ethereum Sepolia (Chain ID: 11155111). All contracts verified on Sourcify.

Contract	Address
AaveV3Adapter	`0x8545D79f6FaB51EDc93Cf024fBD1FfAc98504ba1`
CompoundV3Adapter	`0xEB0D41F07691765314B9A45645Ee995d879c7ac7`
StrategyManager	`0x353469534dA4FB64d52Ae5059CEFd098557eBFa9`
AIVault (proxy)	`0x1324238b6F56Ccc785fC7f79Ca693546236Ad02C`

Tech stack: Solidity 0.8.24, Python 3.12, Foundry, OpenZeppelin, Chainlink, Docker

Design patterns used: ERC-4626 (tokenized vault), UUPS proxy (ERC-1967), Adapter/Strategy pattern, EIP-712 typed signing

Thank You

Questions?

GitHub: `github.com/SergeyySolovyev/ai-yield-vault`

Contact: sesesolovev@edu.hse.ru | @Sergey_Solovjov | www.sergeisolovev.com